

Research

Development of management effectiveness indicators and their evaluation for Cu Lao Cham-Hoi An biosphere reserve in Vietnam

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Abstract

Vietnam has 11 Biosphere Reserves (BRs) designated by UNESCO where evaluation of management effectiveness plays an important role to promote sustainability for these areas. We propose a set of indicators suitable to Vietnam's social-ecological conditions based on the IUCN concepts of management effectiveness and the consultation dialogues with relevant BR stakeholders. This set includes 58 indicators, 20 headline indicators according to 6 management elements of BR: Context, Planning, Inputs, Process, Outputs and Outcomes. Management effectiveness value is divided into 5 levels (0, 25, 50, 75, 100) and thereafter categorised into 4 ranges according to the corresponding value (Very low: 0–25 points; Low: 25–50 points; Average: 50–75 points; High: 75–100 points) with the higher the value, the greater the effectiveness. Based on the indicators, we evaluated the management effectiveness of Cu Lao Cham-Hoi An BR (designated in 2009). The results of evaluating management effectiveness for this BR have determined that the overall management effectiveness for the whole BR is 78.2 out of 100 scores, equivalent to high management effectiveness. Results for management elements: element 1 (Context-85.8 score), element 2 (Planning-83.5 score), element 3 (Inputs-73.4 score), element 4 (Process-76.3 score), element 5 (Outputs-68.3 score) and element 6 (Outcomes-81.9 score). Although the indicators presented a useful analysis of management effectiveness in this particular case, owing the diversity of BRs in Vietnam in terms of socio-ecological conditions, total area, governance model and administration, caution should be applied in the treatment and/or selection of these indicators for all BR cases and the 58 indicators should be used as a guide for each specific BR.

Keywords Biosphere reserves · Management effectiveness indicators · Evaluation · Sustainable development · Vietnam · Cu Lao Cham-Hoi An

1 Introduction

The Biosphere Reserve (BR) is designated areas under the Man and Biosphere Programme (MAB) of the United Nations Educational Scientific and Cultural Organization (UNESCO) as a model to practice sustainable development initiatives or as “learning sites for sustainable development” with three functions (Conservation, Development and

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Logistic support) in three functional zones (Core, Buffer, and Transition), when the core area is legal protected areas (PA) [1–3]. Recent UNESCO policies have promoted BR management for sustainability and response to climate change [4–6]. As of March 2024, the world's BR network has grown rapidly, with 748 sites in 134 countries and territories [7].

The management effectiveness of BR is the degree to which the goals of BR are achieved, expressed according to the three basic functions previously mentioned [2, 8]. The management effectiveness of BRs has mainly focused on successfully implementing their individual functions [9–12], with many prioritizing the relationship between conservation and development [13–16].

The adoption of UNESCO's Seville Strategy and the Legal Framework in 1996 strongly promoted the growth of BRs in the world [17]. Vietnam had its first BR designated by UNESCO in 2000, and to date has eleven BRs representative of ecosystems and diverse administrative management, ethnic groups, populations, and spatial extents [18, 19]. Sustainable management tools are needed to achieve the goals of BRs in functional zones, particularly to improve management effectiveness. Before 2020, UNESCO-designated BRs in Vietnam were mainly managed by local authorities and lacked a unified national management policy [20, 21], legal basis, and resources to perform their functions [20].

A number of indicators to evaluate management effectiveness are used worldwide, such as the Rapid Assessment and Prioritization of Protected Area Management (RAPAM) [22], Management Effectiveness Tracking Tool (METT) [23–26], and the IUCN Green List [27] but these have mainly been for the purposes of assessing the conservation function. The 6-step cycle or elements developed by the IUCN [8, 28] has become a popular approach for evaluating the management effectiveness of the PA's systems. While some have been coincidentally applied to BRs through their application to the PAs that form the core area of BRs [24–26, 29], these have mainly been for the purposes of conservation. For example, the Management Effectiveness Tracking Tool (METT) is based on six management elements (Context, Planning, Inputs, Process, Outputs, Outcomes) with 30 main questions to evaluate PA management effectiveness [26], which continues to be refined as METT-4 with updated 38 questions [29] and has been applied in most countries around the world to evaluate management effectiveness for PAs, including Vietnam [29, 30]. In fact, limited attention has been paid to the management effectiveness of simultaneously achieving the three functions of BRs (conservation, development, and logistical support).

Currently, there are a few sets of indicators for evaluating the management effectiveness of BRs holistically. In order to promote sustainable development for marine and coastal areas, UNESCO has compiled and developed the "Handbook for Measuring the Progress and Outcomes of Integrated Coastal and Ocean Management" [31], which focuses on developing indicators for sustainable management of coastal and marine areas, to achieve the goals concerning: (i) Integrated coastal and ocean management; (ii) Governance performance; and (iii) Ecology. One of the comprehensive criteria for evaluating the management effectiveness of BRs was developed by South Africa [32], which comprises 44 criteria, including 8 general criteria, 7 conservation criteria, 14 sustainable development criteria, and 15 logistics support criteria. Most recently, a new tool, the Biosphere Reserve Effectiveness of Management index (BREMi), was developed and applied on a trial basis for Arab countries to evaluate management effectiveness and support a 10-year Periodic Review [33]. This set included 65 indicators belonging to 34 headline indicators distributed among the six management elements developed by the IUCN [8]. The set of criteria and indicators for evaluating the management effectiveness of BR developed by Son et al. [34], with 38 indicators and 15 criteria according to six management elements, is the most comprehensive in Vietnam. This was experimentally evaluated for the Biosphere Reserves of Cat Ba Archipelago and Cu Lao Cham-Hoi An (CLC-HA).

Several countries, such as Arab countries, have chosen the IUCN management concept [8], which was originally applied to the objective of biodiversity conservation, as a basis for evaluating management effectiveness for BRs while simultaneously implementing three main goals or functions (Conservation, Development and Logistical support) [33]. Therefore, indicators to evaluate the management effectiveness of BRs need to be adjusted to suit the context of the specific BR. Comparing the research on the governance structure of BRs in Japan [35] and the IUCN's management concept [8] has shown similarities between Criterion 2 (Legal frameworks) and element 1 (Context) and element 2 (Planning); Criterion 3 (Administrative institutions) and element 3 (Inputs), especially in human and financial resources; Criterion 5 (Executed BR implementation activities) and element 5 (Outputs) and element 6 (Outcomes); Criterion 3 (Stakeholder identification and decision-making processes) and Criterion 6 (Participatory/collaborative frameworks) and element 4 (Process).

In Vietnam, studies evaluating management effectiveness are still very limited, apart from research evaluating management effectiveness for marine protected areas (MPAs) [30] and evaluating the management effectiveness for Kien Giang BR, using UNESCO tools [9]. The study of indicators to evaluate the effectiveness of BR management proposed by Son et al. [34] is the most complete study in Vietnam.

In 2020, the amended law on Environmental Protection (Law No 72/2020/QH14) [36] officially took effect, identifying the Ministry of Natural Resources and Environment (MONRE) as the national agency responsible for management, with many specific regulations to develop the world's BR system in Vietnam in a sustainable way to perform three functions in three functional zones. Therefore, there is an urgent need to develop a set of indicators to evaluate the management effectiveness of BRs to meet UNESCO's requirements while also being suitable for Vietnam's socio-economic and institutional conditions.

CLC-HA is one of eleven BRs of Vietnam, with coordinates from 15°51'18" to 16°00'00" North latitude and from 108°17'25" to 108°33'30" East longitude, designated by UNESCO in 2009, with a total area of 33,475 ha, of which the core area is the Cu Lao Cham Marine PA. Furthermore, Hoi An Ancient Town was designated as a UNESCO World Cultural Heritage Site in 1999 [37] (See Fig. 1).

This reserve was selected as the study site for several reasons (See Table 1): (1). It represents seven coastal and island BRs out of eleven BRs in Vietnam; (2). It has a more moderate area (33,475 ha) and population (95,227 people) compared to other BRs in Vietnam, suitable for managing and performing their functions well; and (3). The compact administrative management system of the BR belongs to one district; therefore, it is easy to promote cooperation among local communities, local authorities, and other stakeholders. In addition, the CLC-HA BR is a model of sustainable community development in plastic waste management and biodiversity conservation [37, 38], preserving cultural values [39], harmonizing nature conservation and local socio-economic development [38], and scientific research [40–44]. These results provide useful information for evaluating the management effectiveness of the CLC-HA BR based on a set of indicators.

Therefore, to promote the sustainable development of BRs, this study aims to propose a set of indicators to evaluate management effectiveness to best meet Vietnam's management regulations, implement policies of the world's UNESCO/MAB, and use these indicators to evaluate management effectiveness in the CLC-HA BR. This research is also a part of results of the national project "Research on theoretical and practical basis and propose a guiding framework for management of world's biosphere reserves designated by UNESCO in Vietnam."

2 Methods and data

2.1 Development of indicators to evaluate management effectiveness for biosphere reserves in Vietnam

The principles for developing indicators to evaluate the management effectiveness of BRs were as follows: (i) based on functions and consistent with the management objectives of BR; (ii) in accordance with the institutional system and policies of Vietnam and the policies of UNESCO/MAB; (iii) ease of understanding, ease of use, and suitability for local conditions; (iv) accurate and scientific; (v) feasible and practicable; and (vi) consistent with management elements according to the management cycle.

The management effectiveness of a BR was evaluated according to the framework proposed by the IUCN [8], whereby the management process includes six elements (Context, Planning, Inputs, Process, Outputs, and Outcomes), which are evaluated repeatedly to gradually improve management effectiveness. This concept was also applied by many researchers to evaluate the effectiveness of BR's management [32–34]. The characteristics of BR management effectiveness were considered according to management elements; therefore, the indicators must be associated with context, planning, inputs, processes, outputs, and outcomes (Table 2).

Methods to develop indicators to evaluate the management effectiveness of BRs were carried out according to the following steps: (1). Overview and assessment of management effectiveness indicators for relevant areas, especially PAs and BRs worldwide and in Vietnam; (2). Appropriate indicators were selected by evaluating the available indicators to best meet the management objectives of BR under specific conditions in Vietnam; and (3). Consultation with relevant stakeholders, especially the central management agency (i.e. MONRE), Vietnam's MAB Program, the Management Board of BRs, and other relevant stakeholders to adjust and perfect the set of indicators.

The indicators to evaluate management effectiveness using six IUCN elements to serve the implementation of the functions of the BR included (1) those according to Son et al. [34]; (2) those according to Pool-Stanvliet et al. [32]; (3) those according to Stolton et al. [29]; and (4) those according to BREMi [33]; (5) Lima Action Plan (See Table 3). The indicators developed by Son et al. [34] were used as core indicators to propose a new set of indicators because they were the most complete until 2019 for Vietnam's conditions. The promulgation of new regulations (i.e. Environmental Protection Law

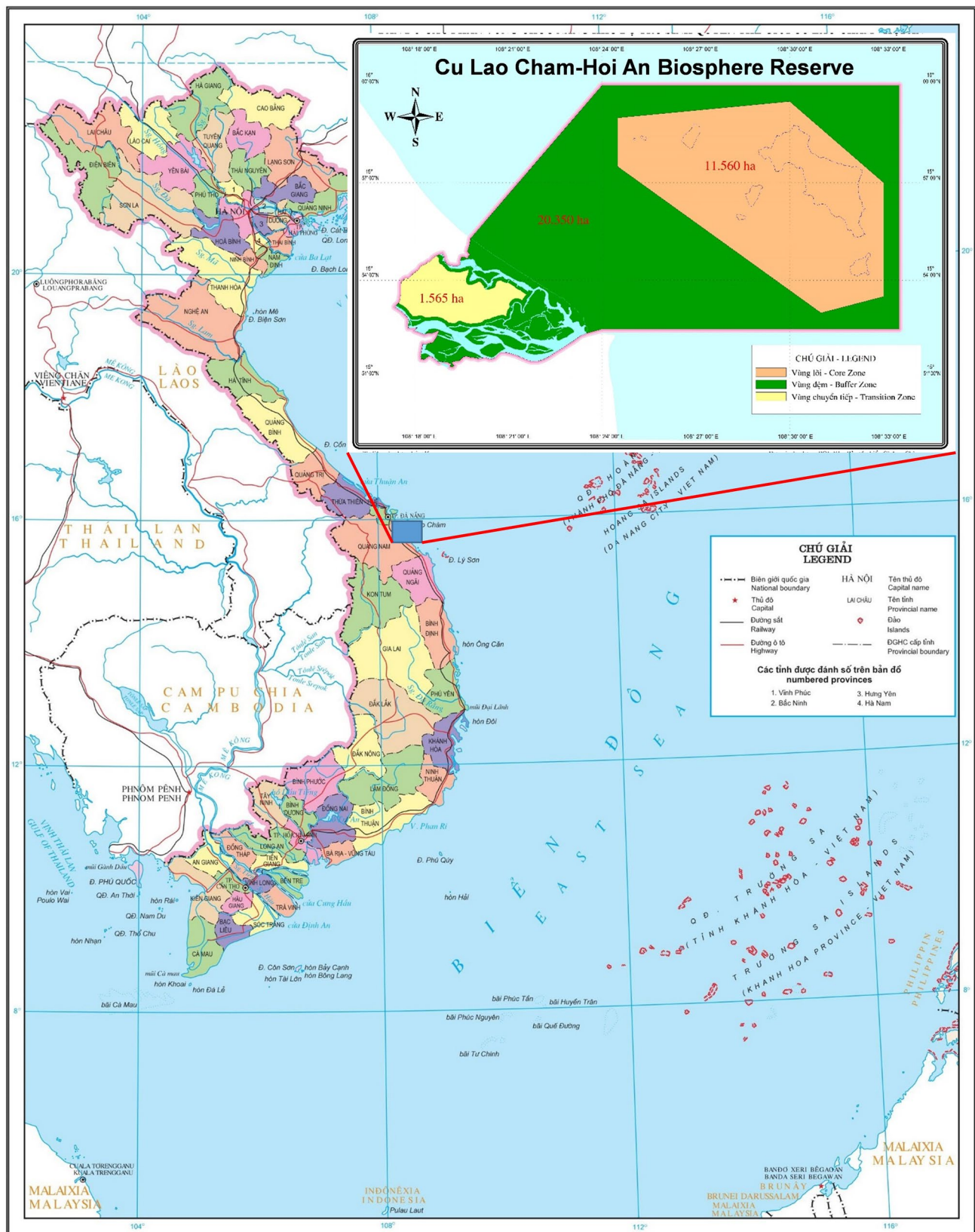


Fig. 1 Location of Cu Lao Cham-Hoi An Biosphere Reserve

Table 1 Summary information about BRs in Vietnam

Biosphere Reserve	Year of designation	Total area (ha)	Population (persons)	Population density (persons/km ²)	Administration
Can Gio Mangrove	2000	75,740	71,157	94	1 district
Cat Ba	2004	26,241	32,000	122	1 district
Red River Delta	2004	105,557	128,075	121	Many districts of many provinces
Kien Giang	2006	1,188,106	353,893	30	Many districts of 1 province
Western Nghe An	2007	1,299,795	918,248	71	Many districts of 1 province
CLC-HA	2009	33,475	95,227	284	1 district
Mui Ca Mau	2009	371,506	170,321	46	Many districts of 1 province
Dong Nai	2011	969,993	485,900	50	Many districts of many provinces
Langbiang	2015	275,439	378,714	137	Many districts of 1 province
Nui Chua	2021	106,646	465,021	436	Many districts of 1 province
Kon Ha Nung Highland	2021	413,511	267,493	65	Many districts of 1 province
Total		4,866,009	3,366,049		
Average		442,373	306,004	69	
Standard deviations		480,035	258,124	122	

2020) and the requirements of national management agencies have created the need to propose a new set of indicators based on existing indicators and proposals from management agencies in Vietnam.

The grouping of indicators was based on the principle of association with six management elements, especially management characteristics for BRs (See Table 2), linking functions, and functional zoning, and was appropriate for Vietnam's conditions. Thus, the indicators were grouped into six large groups (management elements) and subgroups (headline indicators) to easily evaluate management effectiveness. Accordingly, element 1 (Context) included three headline indicators about the current status of management, organization, conservation and impact on biodiversity; element 2 (Planning) included two headline indicators about long-term development strategy and management plan; element 3 (Inputs) included four headline indicators about resources for implementation, inventory and use of resources, and information database; element 4 (Process) included two headline indicators about coordination and cooperation and participation of relevant parties; element 5 (Outputs) included four headline indicators of ecosystem and species trends, and performance of three functions (conservation, development and support), and finally, element 6 (Outcomes) included three headline indicators about connecting functions, network development and summarizing report to gradually improve the BR's management effectiveness.

The indicators to evaluate the BR's management effectiveness were co-developed through a consultative process via three workshops to enlist the participation of the main stakeholders: MONRE, MAB Vietnam, the Management Board of BRs, and other scientists and managers: 1). Workshop "Basic science for sustainable development—sharing experiences of world's BRs and global geoparks in Vietnam," organized by Ministry of Science and Technology (MOST) in cooperation with Ministry of Natural Resources and Environment (MONRE) and Vietnam's Man and Biosphere Program National Committee (MAB Vietnam), in Hanoi, August 11, 2022; and 2). Two scientific workshops in the framework of the project "Research on the theoretical and practical basis and propose a guiding framework for management of world biosphere reserves recognized by UNESCO in Vietnam" (code DTDL.XH-06/21), funded by MOST, held in Hanoi on November 25, 2022, and December 23, 2023. In addition, these indicators were consulted in a meeting with the staff of MONRE in Hanoi on April 26, 2023. We obtained the written approval for publication from all the relevant participants and stakeholders in the interviews and workshops organized by MOST in cooperation with MONRE and MAB Vietnam.

Comments on these indicators from relevant stakeholders are summarized as follows: (1). Agree on the importance of building indicators to evaluate the management effectiveness of BRs in Vietnam. (2). Ensure the feasibility of the indicators using the available data; (3). Linked with six management elements and BR functions; (4). Meet the diverse requirements of BRs in terms of spatial scale, population, and administrative management; (5). Effectiveness evaluation indicators must be linked to the characteristics of the six management elements of BRs (Tables 2) and (6). The evaluation methods for these indicators are easy to implement and feasible when considering the shortage of relevant documents and databases on BRs.

Table 2 Characteristics of management effectiveness according to management elements for BRs

No	Management elements	Characteristics according to management elements
1	Context	- The management apparatus must be consistent with BR's development - The organizational structure, including resources in core areas, buffer zones and transition areas, must be consistent with the implementation of BR's functions - Zoning and functions of the zones and their boundaries must be clearly defined - Threats to biodiversity must be identified
2	Planning	- Long-term vision, goals and objectives are clearly defined in BR's management regulations, aiming to simultaneously implement 3 its functions - BR's development plans (medium and short term) must meet its functions and functional zones - Socio-economic and sectorial development strategies, planning and plans must be integrated with BR's development goals - BR's functional zoning must be consistent with provincial planning (up to 2030 and beyond) and sectorial planning
3	Inputs	Human, financial and material resources and database must ensure quantity and quality to properly implement BR's management plans
4	Process	Process and coordination mechanism of BR's Management Board vertically (province/city level, functional zones, locality (district, commune)), horizontally (core areas, buffer zones, transition areas) and between relevant parties must be developed and implemented
5	Outputs	Output products and services must be associated with the implementation of 3 functions in the entire BR as well as in functional zones in the direction that biodiversity must be conserved, development in harmony with nature and research & education for the sake of sustainable development goals
6	Outcomes	BR's functions must be performed effectively, BR's reputation and brand of the Biosphere Reserve must be enhanced, and solutions to improve BR's management effectiveness must be proposed and implemented

Table 3 Characteristics of the referenced existing indicators of management effectiveness evaluation

Sources of indicators	Management principles	Number of indicators	Place of application
Son et al. [34]	IUCN's 6 elements	38	BRs in Vietnam
Pool-Stanvliet et al. [32]	Functions of BR	44 (General 8; Conservation 7; SD 14; Support 15)	BRs in South Africa
Stolton et al. [29]	IUCN's 6 elements	38	PAs worldwide
BREMI [33]	IUCN's 6 elements	65	BRs in Arab countries
Lima Action Plan [6]	Sustainable development		BRs worldwide

Therefore, this set of indicators is considered the core indicator for evaluating management effectiveness, and each BR depends on its specific conditions to supplement the other indicators accordingly.

The proposed indicators were evaluated according to scores, ranging from 0 points (min value) to 100 points (max value) using Scoring Instructions, which cover titles such as *No*; *Headline indicator*; *Indicator*; *Measurement unit*; *Evaluation method*; *Documents and supporting data*; and *Meaning of indicator* (See Scoring Instructions in Annex 1) to evaluate the effectiveness of management of BR in Vietnam according to points: 0-25-50-75-100. The scores of the headline indicators were calculated using the average value of the corresponding indicators, and the scores of the component indices according to the management elements were calculated using the average value of the corresponding headline indicators. The higher the score, the higher is the management effectiveness according to the corresponding indicator or index. Management effectiveness is divided into 5 main levels (0-25-50-75-100), for easy classification in the evaluation (See Annex 1). However, in practice, due to lack of data or supporting information, the evaluation score may fall between the above 2 levels (for example between 50 and 75), then the evaluator can give his subjective score according to the 5-point interval (ie 50-55-60-65-70-75). This scoring allows the evaluator to give the score closest to their experience. Management effectiveness is also categorised into 4 ranges according to the corresponding value (Very low: 0 to less than 25 points; Low: 25 to less than 50 points; Average: 50 to less than 75 points; High: 75 to 100 points).

The indicators to evaluate management effectiveness are to help the BR's Management Board and local provincial governments in Vietnam develop a tool to evaluate management effectiveness towards sustainability. Depending on their needs and available resources, they will be the ones to conduct a self-assessment of the management effectiveness of the site, based on the indicators proposed by this study. Vietnam's Man and Biosphere National Committee can provide consultation and technical support for evaluating the management effectiveness of specific BRs based on their requirements.

2.2 Evaluation of management effectiveness for Cu Lao Cham-Hoi An Biosphere Reserve

The indicators proposed above were applied to evaluate the management effectiveness of CLC-HA BR. The evaluation steps were as follows: (1). A team of evaluators was established, including external experts and representatives of the BR's Management Board, which are in this study and are the authors of this paper and members of the Secretariat of the CLC-HA BR; (2). The evaluation team collected, synthesized, and analyzed existing documents related to the content of all evaluation indicators with the aim of identifying evidence for scoring; (3). This team then conducted a collective evaluation and scores each indicator according to scoring instructions using qualitative and quantitative methods; (4). Analyzed evaluation results according to indicators, headline indicators, and component indices according to management elements using descriptive statistical analysis methods (such as mean value and standard deviation); and (5). Provided comments on management effectiveness according to indicators, headline indicators, and indices for management elements to provide a basis for proposals to improve management effectiveness for the management boards of BRs and relevant parties.

The evaluation process was carried out from 17 to 21 December 2023, at the CLC-HA BR's Management Board, Hoi An Town, Quang Nam Province.

As the evaluation of management effectiveness is mainly based on available information, the data collection process is extremely important. The main data collected included the following: (1). Policies related to BR's functions (conservation, sustainable development, and support) in Hoi An City and Quang Nam Province; (2). Management regulations, strategies, development plans, and functional zoning of CLC-HA BR; (3). Reports related to organizational structure,

human resources, finance, and facilities serving BR's management; (4). Nomination reports and 10-year review reports on BRs; (5). Annual and 5-year management reports for BR; (6). Reports on the programs and projects implemented on BR related to 3 functions; (7). Socio-economic, biodiversity, and environmental data related to the BR; and (8). Functional zoning and thematic maps related to the BR.

These documents, data, and reports were then classified, synthesized, and analyzed based on the Scoring Instructions described in Annex 1 to easily perform the evaluation.

Following the evaluation steps and data availability, the team conducted a collective evaluation and scored each indicator according to scoring instructions using qualitative and quantitative methods. If the data/information or documents were completed, systematic, related to the indicator and meet the requirements in the scoring guide (i.e. Measurement unit), then the "quantitative" or "quasi-quantitative method" could be used to score. In many cases, data/information/documents were insufficient; therefore, the evaluation team must rely on subjective assessments or other relevant information to evaluate according to the range (i.e., Very low, Low, Medium, Average, High) to agree on scoring. In that case, a "qualitative method" was used. The practical experience of the BR Management Board was particularly meaningful in this evaluation. Furthermore, by conducting this evaluation, the Management Board can compile a suitable database for further evaluation.

Management effectiveness evaluation and scoring according to indicators were conducted mainly by the self-assessment principle of BR's Management Board, based on available data and evidence, and using the scoring instructions (Annex 1 and 2). The compiled documents and data (i.e., supporting documents) are compared with the scoring method (i.e., measurement unit) to determine the score for each indicator (See Table 4), as the practical experience of the BR's Management Board was especially meaningful in this evaluation. Owing to the limitations in the BR database, the evaluation mostly used qualitative methods partly combined with quantitative methods when sufficient data were available.

3 Results

3.1 Indicators of management effectiveness evaluation for biosphere reserves in Vietnam

By reviewing existing indicators to evaluate the management effectiveness of BRs worldwide and in Vietnam, and through the consultation process with key stakeholders related to BRs, such as MONRE, Biosphere Reserve Management Boards, Vietnam's MAB, scientists, and managers, a set of indicators to evaluate management effectiveness is proposed. The set of indicators used to evaluate Vietnam's BR management effectiveness included 58 indicators belonging to 20 headline indicators according to six management elements (See Appendix 1). This set of indicators was compiled from the following sources: (1). Set of criteria and indicators proposed by Son et al. [34]; (2). Indicators developed in South Africa [32]; (3). Indicators according to METT-4 [29]; (4). According to BREMi [33] and (5). According to a proposal from the Department of Nature Conservation and Biodiversity, MONRE which is the National Management Agency for BRs in Vietnam. These proposed indicators are also related to the Lima Action Plans (LAP) [6] to assess the implementation of UNESCO's BR policy.

Thus, the number of indicators is distributed by management elements as follows: (1) Context: eleven indicators, four headline indicators; (2) Planning: five indicators, two headline indicators; (3) Inputs: eleven indicators, four headline indicators; (4) Process: five indicators, two headline indicators; (5) Outputs: sixteen indicators, five headline indicators; and (6) Outcomes: eight indicators and three headline indicators.

In principle, BRs in Vietnam can refer to and apply these indicators to evaluate and monitor management effectiveness.

3.2 Management effectiveness of Cu Lao Cham—Hoi an biosphere reserve

The CLC-HA BR was selected for experimental evaluation of the proposed management effectiveness indicators. The evaluation of management effectiveness, including 58 indicators according to 20 headline indicators in six management elements, was conducted with scores ranging from 45 to 95 points (Table 5).

Table 4 Example of evaluation and scoring management effectiveness according to indicators for CLC-HA BR

No	Headline indicator	Indicator	Measurement unit	Supporting documents	Analysis and evaluation	Score
1.	CONTEXT					
1	1.1. <i>Current status of management and organization</i>	1.1.1. Organizational structure of Biosphere Reserve	-No appropriate structure or organization: 0 points -Having structure & organization, but less appropriate in performing BR's functions: 25 points -Having a fairly appropriate structure & organization in performing BR's functions: 50 points -Having appropriate structure & organization in performing BR's functions: 75 points -Having a very appropriate structure & organization in performing BR's functions: 100 points	- Regulations on management of CLC-HA BR (2015) - Functions and tasks of the BR's Management Board (2015) - Decision to strengthen the annual Management Board's personnel (2021) - Organizational chart of the Management Board (2021)	- The structure of the BR is very suitable in performing its functions, with the Management Board representing leaders of departments and branches, especially leader of Marine PA (core area) is Standing Deputy Chairman of the Board, so resources can be mobilized to perform well the BR's functions; - The Secretariat has an advisory and coordinating role, with the assistance of the Technical Team; - The advisory board includes managers, experts and community representatives;	95

Table 5 Indicators and headline indicators to evaluate management effectiveness according to management elements for CLC-HA BR

Indices according management element (score and standard deviation)	Headline indicators (score and standard deviation) (points)	Indicators (score) (points)
1. Context: 85.8 (10.03)	1.1. Current status of management and organization: 95.0 (0)	1.1.1. Organizational structure of BR: 95 1.1.2. Management regulations of BR: 95 1.1.3. Operational regulations of the BR Management Board: 95 1.2.1. Boundaries of zones: 90 1.2.2. Functions of zones: 90 1.3.1. Conservation status of endangered, precious and rare species: 80 1.3.2. Conservation status of typical habitats: 90 1.3.3. Conservation status of ecosystems: 90 1.4.1. Level of threat to typical ecosystems: 65 1.4.2. Level of threat to endangered, precious and rare species: 65 1.4.3. Trending impact of invasive alien species: 85
2. Planning: 83.5 (0.35)	2.1. Long-term vision for sustainable development of BR: 83.8 (8.54)	2.1.1. Developing development strategy of BR: 95 2.1.2. Implementation of the community development program of the strategy: 85 2.1.3. Implementation of the science and technology program of the strategy: 75 2.1.4. Implementation of the education, awareness and communication programs of the strategy: 80 2.2.1. Mid-term (5-year) management plan of BR: 90 2.2.2. Annual management plan BR: 85 2.2.3. Local sustainable development plan: 75
3. Inputs: 73.4 (7.86)	3.1. Mobilizing resources to implement the BR's management plan: 71.3 (4.79)	3.1.1. Human resources to implement BR's management plan: 75 3.1.2. Equipment to implement BR's management plan: 65 3.1.3. Financial resources to BR's management plan: 70 3.1.4. Information and database for implementing BR's management plan: 75 3.2.1. Natural Resources/Biodiversity Inventory: 75 3.2.2. Inventory of human values: 95 3.3.1. Sustainable use of ecosystem services: 75 3.3.2. Sustainable use of local genetic resources in BR: 60 3.4.1. Development of information center to serve education and tourism: 70 3.4.2. Development of agricultural, forestry and fishery products with BR's brand: 70 3.4.3. Development of tourism products and services with BR's brand: 70

Table 5 (continued)

Indices according management element (score and standard deviation) (points)	Headline indicators (score and standard deviation) (points)	Indicators (score) (points)
4. Process: 76.3 (12.37)	4.1. Coordination and cooperation mechanism: 67.5 (3.54)	4.1.1. Coordination and cooperation mechanism in BR: 65 4.1.2. Coordination mechanism for inter-commune, inter-district and inter-provincial cooperation in BR: 70
	4.2. Participation of relevant parties: 85.0 (8.66)	4.2.1. Participation of local communities in BR's management: 95 4.2.2. Participation of business enterprises: 80 4.2.3. Participation of research institutes and universities: 80
5. Outputs: 68.3 (10.92)	5.1. Change trends of typical ecosystems species of BR: 50.0 (0)	5.1.1. Ecosystem change trend in BR: 50 5.1.2. Change trends of endangered, precious and rare species in BR: 50
	5.2. Implementation of conservation function and sustainable use of natural resources: 77.5 (5.00)	5.2.1. Project, program on ecosystem conservation and restoration: 75 5.2.2. Project, program on species conservation and restoration: 75 5.2.3. Project, program on invasive alien species control: 75 5.2.4. Project program on biodiversity inventory and monitoring: 85 5.3.1. Sustainable livelihood development: 90 5.3.2. Development of organic agriculture: 80 5.3.3. Development of environmentally friendly industries and handicrafts: 50
	5.3. Implementation of development functions: 71.7 (19.15)	5.3.4. Development of green and sustainable ecotourism: 80 5.3.5. Effective works/measures to prevent pollution, environmental degradation and natural disasters: 45 5.3.6. Maintenance and development of cultural values and traditions of BR: 85
	5.4. Implementation of logistic support functions: 75.0 (21.21)	5.4.1. Research on Biosphere Reserve: 60 5.4.2. Education and awareness raising about Biosphere Reserve: 90
	5.5. Social responsibility of business enterprises: 67.5 (10.61)	5.5.1. Number of social enterprises participating in implementation of BR's functions: 75 5.5.2. Programs led and invested by businesses enterprises: 60

Table 5 (continued)

Indices according management element (score and standard deviation) (points)	Headline indicators (score and standard deviation) (points)	Indicators (score) (points)
6. Outcomes: 81.9 (4.61)	6.1. Connecting functions in BR: 81.7 (2.89)	6.1.1. Connecting support function with biodiversity conservation function: 85 6.1.2. Connecting support function with the development function: 85 6.1.3. Connecting conservation function with the development function through linking ecosystems of core areas: 80 6.2.1. Cooperation in BR's network: 75 6.2.2. Promotion of BR's title/brand: 80 6.3.1. Summary report of BR's Management Board: 80 6.3.2. Using monitoring and evaluation results to improve management effectiveness: 90 6.3.3. Organizing monitoring and evaluation of sustainable development of buffer zones and transition areas: 90 58 indicators
6 Management elements: 78.2	6.2. Contribution to BR's network activities: 77.5 (3.54) 6.3. Summary of management activities and improvement of BR's management effectiveness: 86.7 (5.77) 20 headline indicators	

3.2.1 Evaluation of the management effectiveness of the entire CLC-HA BR

The aggregate management effectiveness value for the entire BR is 78.2 points, which is equivalent to high management effectiveness in a lower range (high management effectiveness varies from 75 to 100 points).

3.2.2 Evaluation of management effectiveness according to management elements

The management effectiveness values by element are presented in Table 5 and Fig. 2. The management effectiveness of elements 1 (Context), 2 (Planning), 4 (Process), and 6 (Outcomes) have high scores of 85.8, 83.5, 76.3 and 81.9 points, respectively; therefore, these elements are maintained. However, element 1 had a high standard deviation value of 10.03 points, indicating a large degree of variation between the heading indicators, with indicator 1.4 (Threats or pressure on biodiversity) having the lowest value, 71.7 points.

In contrast, element 3 (Inputs) and element 5 (Outputs) have lower value of 73.4 and 68.3 points, meaning they need to be improved to make management effective for the entire BR increase. In particular, element 5 has a high standard deviation value, 10.92 points with large range (from 50.0 to 77.5 points) when headline indicator 5.1. (Change trends of typical ecosystems and species of the BR) highlights the challenge of biodiversity degradation in the future and the need to prioritize improving the situation.

3.2.3 Evaluation of management effectiveness according to headline indicators

The value of management effectiveness according to the headline indicators is shown in Table 5 and Fig. 3. Accordingly, twelve out of twenty headline indicators (60.0%) have a value of 75 points or more, equivalent to high management effectiveness and most of them belong to element “Context” (4), element “Planning” (2) and element “Outcomes” (3) (Table 6). Therefore, the headline indicators should be maintained. In contrast, several headline indicators, such as trends in ecosystem and species change (5.1:50.0), use of biodiversity (3.3:67.5 points) and social responsibility of businesses (5.5:67.5 points) need to be prioritized for improvement in the future.

3.2.4 Evaluation of management effectiveness according to indicators

The value of management effectiveness according to the indicators is shown in Fig. 4. Accordingly, 42 of the 58 indicators (72.4%) had a value of 75 points or more, which is equivalent to high management effectiveness, and they should be maintained. Especially, the management issues related to the indicators with the lowest value, such as 5.1.1 (Ecosystem change trend in BRs: 50 points), 5.1.2 (Change trends of endangered, precious and rare species in BRs: 50 points), 5.3.3 (Development of environmentally friendly industries and handicrafts: 50 points); and 5.3.5 (Effective works/measures to

Fig. 2 Management effectiveness of CLC-HA BR according to management elements

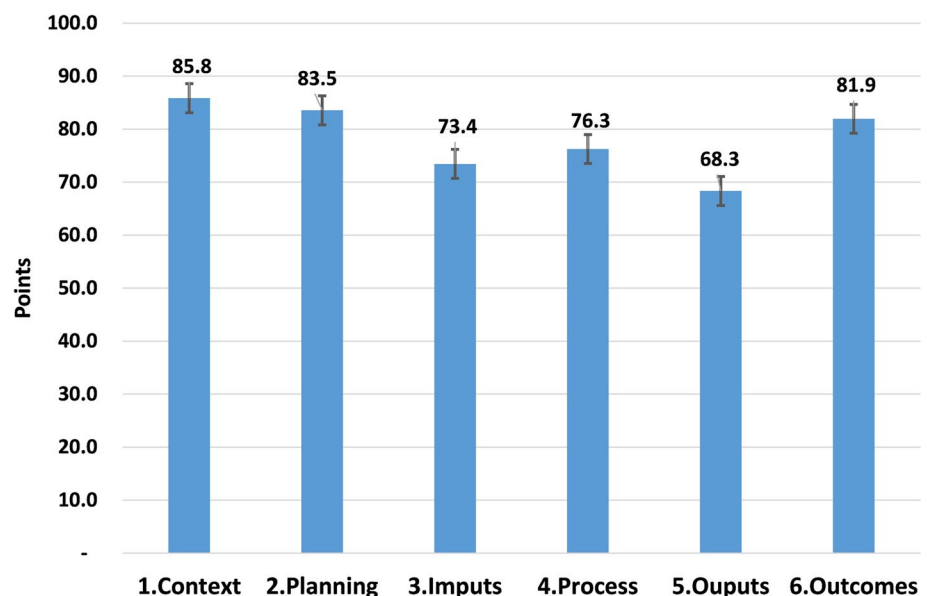


Fig. 3 Management effectiveness of CLC-HA BR according to headline indicators

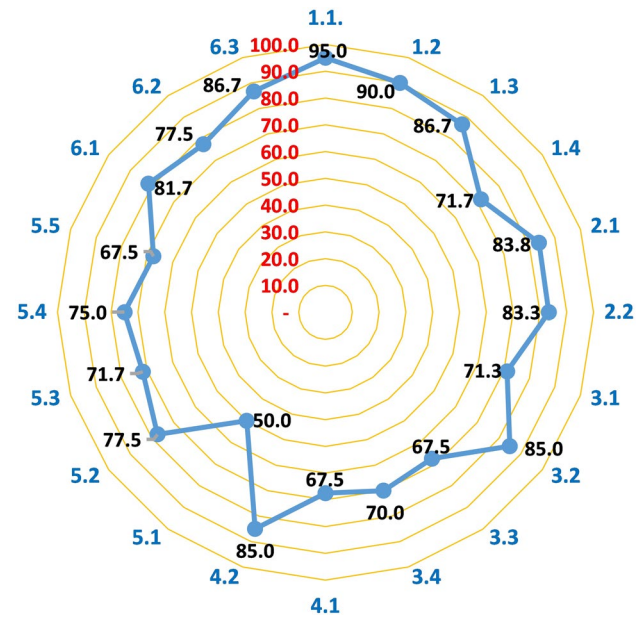


Table 6 Comparison of management effectiveness assessment for Biosphere Reserves according to management elements (Score)

No	Management elements	CLC-HA BR (in 2023)	CLC-HA BR (in 2018) [34]	Arab RBs (in 2022)) [33]
1	1.Context	85.8/100	3.0/4 (70.0/100)	6.9/10 (69/100)
2	2.Planning	83.5/100	3.25/4 (81.25/100)	7.0/10 (70/100)
3	3.Inputs	73.4/100	2.375/4 (59.36/100)	4.9/10 (49/100)
4	4.Process	76.3/100	3.0/4 (75.00/100)	6.3/10 (63/100)
5	5.Outputs	68.3/100	2.5/4 (62.50/100)	5.1/10 (51/100)
6	6.Outcomes	81.9/100	3.125/4 (78.13/100)	6.8/10 (68/100)
	Overall	78.2/100	2.875/4 (71.88/100)	6.2/10 (62/100)

prevent pollution, environmental degradation and natural disasters: 45 points) need to be resolved in order to gradually improve management effectiveness for the entire BR. In addition, it is necessary to pay attention to and improve management issues related to the sustainable use of local genetic resources (3.3.2:60 points) and Research on BRs (5.4.1:60 points).

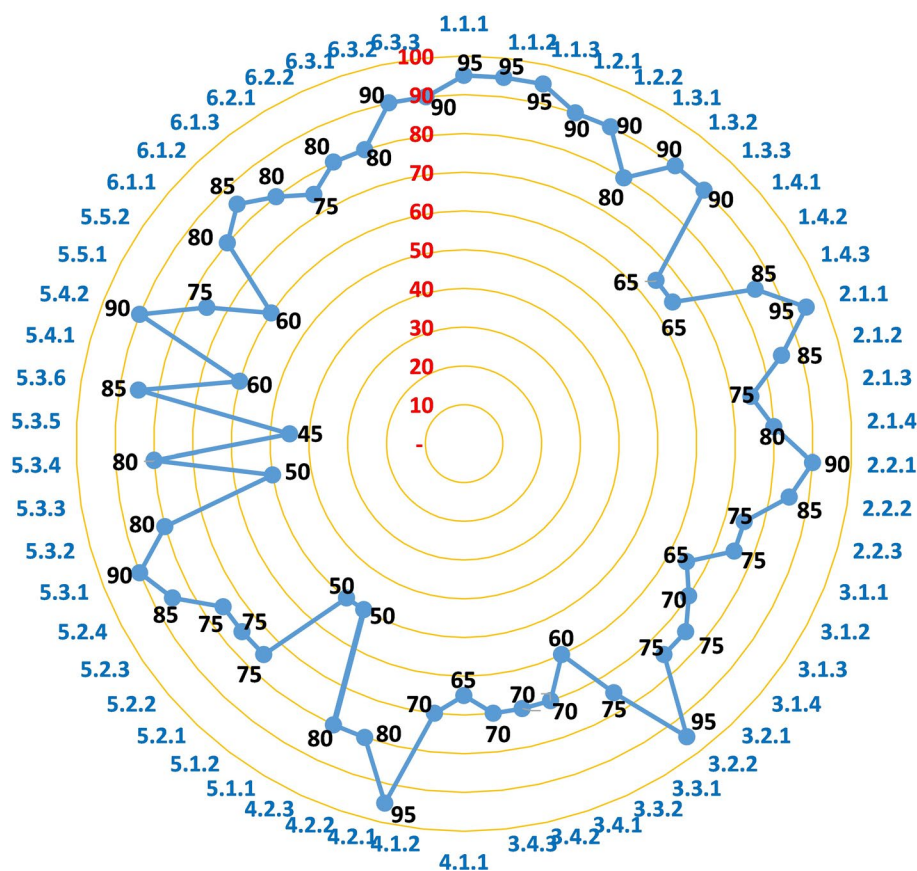
The characteristics of management effectiveness according to the management elements for the CLC-HA BR are described in the appendix 2.

The effectiveness of CLC-HA BR's management according to "Context" with a high evaluation value (85.8 points) has the following characteristics: (1). A strong management board representative of the departments and branches of Hoi An City, supported by a diverse and experienced technical team and an advisory board, has developed appropriate management regulations and functional zoning to fully implement its functions; (2). Research on ecosystems, habitats, and rare and endangered species, especially in core areas, has been conducted well, meeting the requirements of biodiversity conservation; and (3). The threats to ecosystems and rare plant and animal species remain at a small to moderate level.

Management effectiveness according to "Planning" with a high evaluation value (83.5 points) has the following characteristics: (1). BR has built a strategic vision for the sustainable development of BRs (BR's Sustainable Development Project) and laid the foundation for developing medium- and short-term plans to fully implement its functions; (2). The basic contents of BR's Sustainable Development Project, such as community development programs, science and technology programs, education programs, awareness raising, and communication, have been implemented relatively well, especially with the active participation of the local community; (3). Based on the sustainable development project, medium-term and annual management plans were developed to perform BR's functions; and 4). The sustainable development plan (action plan to implement the 2030 Agenda for sustainable development and plan to implement the green growth strategy) was developed at the provincial level, which is the guiding framework for the sustainable development of the BR.

Management effectiveness according to "Inputs" with an average evaluation value (73.4 points) has the following characteristics: (1). Human resources and information centers with databases are good at implementing management

Fig. 4 Management effectiveness of CLC-HA BR according to indicators



goals, whereas equipment, facilities, and financial resources are still limited; (2). The monitoring programs were conducted well, especially in terms of cultural and humanistic values, because Hoi An City is a UNESCO Cultural World Heritage; and (3). Developing information centers and building products bearing the BR brand have been done well, contributing to promoting CLC-HA BR at home and abroad.

Management effectiveness according to “Process” with a high evaluation value (76.3 points) has the following characteristics: (1). Although coordination and cooperation mechanisms between relevant parties and functional areas have been clearly defined in BR’s management regulations, in reality, they are only implemented well in the core area and not yet strictly for buffer zones and transition area; and (2). The participation of relevant parties is quite high, especially the local community of the core area, in planning, implementation, and monitoring results, while the participation of businesses and the scientific community is lower because of the shortage of necessary resources.

Management effectiveness according to “Outputs” with an average evaluation value (68.3 points) has the following characteristics: (1). Ecosystems and rare and endangered species in the BR are likely to remain the same or slightly decrease; (2). Programs and projects on the conservation and restoration of ecosystems and species, prevention of invasive alien species, and biodiversity monitoring have been implemented quite well, especially in the BR’s core area; (3). Implementing development functions, such as developing sustainable livelihoods, promoting organic agriculture, green ecotourism, and maintaining and developing cultural and traditional values, has been done well; while, developing environmentally friendly industries, handicrafts, and projects to prevent pollution, environmental degradation, and natural disasters face many difficulties; (4). Regarding the support function, education and awareness-raising activities are conducted well, especially with the participation of local communities, whereas scientific research activities are limited because of a shortage of necessary resources; and (5). Business participation is not significant, mainly in environmentally friendly production and business activities, whereas the number of programs and projects led or invested in by businesses remains limited.

Management effectiveness according to “Outcomes” with high evaluation value (81.7 points) has the following characteristics: (1). Projects with the goal of connecting the BR’s functions (connecting the support function with the conservation function and connecting the conservation function with the development function) are being implemented

with good results; (2). BR's contribution to the domestic and international network is relatively good through cooperative activities and the promotion of BR's brand building; and (3). Implementing summary reports, using assessment results, and monitoring and evaluating sustainable development in buffer zones and transitional areas also contribute significantly to gradually improving management effectiveness.

4 Discussion

The development and testing of indicators for the CLC-HA BR plays an important role in evaluating the effectiveness of BR management in Vietnam. The use of management effectiveness indicators can be applied to different scopes of BR. A composite index (78.2 points) evaluates the effectiveness for the entire BR. Six management effectiveness indicators evaluate for 6 management elements (85.8, 83.5, 73.4, 76.3, 68.3, 81.9 points respectively) of BR. 20 headline indicators and 58 specific indicators evaluate management effectiveness according to more detailed contents or aspects of management elements. Evaluating management effectiveness according to these scopes allows managers to know the strengths or achievements of their BR management to continue maintaining it, but more importantly, to know the limitations, difficulties, or challenges of management activities to provide priority solutions to gradually improve management effectiveness. Furthermore, each BR can apply these indicators in certain periods (for example, 5 years, 10 years) to determine the level of progress in their management activities, particularly relevant for those BRs in their preparations for the periodic review process.

Due to the diversity in area, population, administrative management, and subjectivity in evaluating management effectiveness (i.e., largely self-assessed by the BR's Management Board), as well as the available database, the comparison results between BRs applying these indicators may be relative. However, detailed evaluations at different scopes (the entire BR, by elements, headline indicators, and detailed indicators) can provide an important basis for improving BR management effectiveness.

Although different in purpose, scope, content, and number of indicators, comparing the results of evaluating management effectiveness between the CLC-HA BR and other BRs also has value when only a single composite indicator or index can be used to evaluate management effectiveness for the entire BR. Thus, comparing the evaluation results for the CLC-HA BR in 2023 with 58 indicators and those in 2018 with 36 indicators [34] shows certain similarities with a significant improvement in overall management effectiveness when this effectiveness increases from 71.9 points to 78.2 points over a period of five years. The value of evaluating the management effectiveness of the CLC-HA BR is also compatible with the assessment of BRs in Arab countries using the BREMi Index [33], with a value of 62 points out of 100 (See Table 6). Thus, improving individual BR management will also undoubtedly ultimately improve the implementation success of the Vietnamese MAB programme as a collective.

Among the management elements, the elements "Context," "Planning" and "Outcomes" often achieve higher results in the case of Cu Lao Cham – Hoi An and Arab country BRs while "Inputs" such as human, financial and material/infrastructure and "Outputs" as specific management products and services in implementing BR's functions achieving lower scores is something that is also common to BRs elsewhere across the WNBR [20, 48] that note similar challenges. These facts also point out the challenges in BR's management works in many countries.

The evaluation results for CLC-HA BR for 2018 and 2023 also indicate a significant improvement in management effectiveness in "Context," "Inputs" and "Outputs," and the situation remains the same or slightly improves for "Planning," "Process" and "Outcomes."

However, using these indicators to evaluate management effectiveness also has limitations when qualitative methods are used, and the scoring results depend on the evaluator's subjectivity. Quantitative or semi-quantitative methods can only be used when there are comprehensive and systematic data; however, in reality, the problem of information storage and databases is a big challenge for BRs in Vietnam. Therefore, for future studies, applying the Delphi approach [49] can also reduce subjectivity in this research.

To overcome the challenges in evaluating management effectiveness among BRs with socio-ecological diversity, one proposal is to divide BRs into three subgroups according to the level of complexity in governance (at the same time, corresponding to the spatial scope), as follows: (1). The BRs belong to one district in (3), (2). The BRs belong to many provincial districts (6), and (3). The BRs belong to several districts in several provinces (2). Biosphere reserves within each subgroup can be used to more easily compare management effectiveness. Comparing management effectiveness between BRs of different subgroups may not be accurate, so more research and time are needed to complete the evaluation indicators.

It is also important to note that the indicators used to evaluate Vietnam's management effectiveness were initially linked to UNESCO's LAP (2016–2025), focusing mainly on Strategic Action Area A (The World Network of BRs consists of effectively functioning models for sustainable development) (See Table 5), as these indicators are often applied at the sub-regional and provincial levels for Vietnam's conditions. Only a few indicators relate to Strategic Action Area B on cooperation and global networks and Strategic Action Area C on external partnerships and sustainable funding, as these areas are conducted mainly at an international scale. Finally, this set of indicators has been developed to meet new requirements for the BR's management policies in the specific context of Vietnam [41, 45–47], especially in proposing indicators for planning and mobilizing input resources, processes, and outputs, while still complying with major UNESCO/MAB policies.

Thus, managers of BR can have a comprehensive view of the management effectiveness of the entire BR when considering evaluation values according to the indices of management elements and headline indicators to develop appropriate strategies for sustainable development. Technical staff can clearly see good results in promoting and managing issues to overcome when considering the evaluation value according to the headline and specific indicators. In summary, these indices and indicators can be used as tools to review and evaluate the management effectiveness of BRs in Vietnam.

In short, management effectiveness evaluation indicators are useful tools for evaluating the effectiveness of BR management in Vietnam, and can be referred to by other BRs worldwide. In Vietnam, this is also considered a tool for the BR's Management Board to self-review management work at all levels, according to management elements, headline indicator content, or specific indicators, to gradually improve management effectiveness. However, these indicators need to continue to be evaluated in many BRs of Vietnam to complete, as well as promote the sustainability of BRs by improving management effectiveness.

5 Conclusion

A set of specific 58 indicators, 20 headline indicators regarding the six management elements to evaluate BR's management effectiveness was developed based on a review of existing management effectiveness indicators worldwide and in Vietnam in the context of Vietnam's Environmental Protection Law taking effect in 2020 with many specific regulations on BRs. These indicators have been experimentally evaluated for the CLC-HA BR in 2023 which has achieved results of overall high management effectiveness. This has also been improved compared to the evaluation conducted in 2018. This set of indicators can be used by the BR's Management Board and local authorities as a tool to evaluate management effectiveness through periods with the cautions that careful contextual selection of indicators should be made on a case-by-case basis, using the 58 indicators as a guide for what will 'work' in each BR's specific case. These indicators should be applied for other BRs in Vietnam to continue to complete for coming time.

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Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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